Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/05/2024
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	xmao@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8373
Organization Name:	Loma Linda University		
PI Address 1:	Radiation Medicine		
PI Address 2:	11175 Campus Street, Rm A1010, Chan Shun Pavilion		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	06/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Klotz, Rebecca	**Contact Phone:**	650-604-1119
Contact Email:	rebecca.a.klotz@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 06/30/2025 per NSSC information (Ed., 9/2/24) NOTE: End date changed to 10/31/2024 per NSSC information (Ed., 11/9/23) NOTE: End date changed to 10/31/2023 per F. Hernandez/ARC (Ed., 2/27/23) NOTE: End date changed to 10/31/2022 per NSSC (Ed., 12/14/21) NOTE: End date changed to 10/31/2021 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			

COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)
Grant/Contract No.:	NNX15AB41G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study from space flight mission to ISS will provide a unique opportunity to evaluate the effects of space hazards and mechanisms of response and remodeling of the ocular system. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	Astronauts returning from spaceflight have experienced eye problems, which may decrease retinal performance and lead to long-term effects on visual acuity. This study leverages the collected data from spaceflown murine retinas that were treated with redox-active metalloporphyrin (BuOE) to mitigate spaceflight-induced changes and respective ground controls. 10-week-old adult C57BL/6 male mice (n=5 in each of BuOE treated and saline control groups for spaceflown and ground control samples) were flown on SpaceX-24 Rodent Research-18 (RR-18) to the International Space Station (ISS) national lab, kept in low Earth orbit for 35 days, and returned to Earth alive. Our multi-omics analysis of RNA sequencing (RNA-Seq) and reduced representation bisulfite sequencing (RRBS) data, generated from subsequent murine retina tissues, uncovered genes, pathways, and epigenetic modifications consistent with therapeutic potential of BuOE. From RNA-Seq analysis of spaceflown murine samples, the treatment group show differentially expressed genes relative to saline controls that reached significance (adjusted p-value < 0.05) and included genes Gpx3 and Crhbp, which are related to protection against cell oxidative damage and cellular response to organonitrogen compounds. Ranked fold-changes from the same contrast were used for gene set enrichment analysis, which showed biological processes reaching significance (adjusted p-value < 0.05), including glutathione metabolic processes and cellular response to xenobiotic stimulus. RRBS data of the spaceflown murine samples found 139 hyper or hypo differentially methylated sites spread across chromosomes 1-19 (20% promoters, 21% exons, 43% introns | 20 CpG islands, 7 CpG shores) with a 10% methylation difference (q-value < 0.05). The findings from this investigation have the potential to provide valuable insights into the molecular mechanisms underlying conditions like spaceflight associated neuro-ocular syndrome and assess the effectiveness of BuOE as a countermeasure for astronauts experiencing neuro-ophthalmic abnormalities, which can lead to long-term effects on visual acuity.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW. "Impacts of space radiation and spaceflight on neurovascular remodeling and blood-brain barrier integrity." 22nd Annual NeuroTech Convention, Los Angeles, CA, March 13-18, 2024. Abstracts. 22nd Annual NeuroTech Convention, Los Angeles, CA, March 13-18, 2024. , Mar-2024
Abstracts for Journals and Proceedings	Stanbouly S, Mao XW. "Effects of spaceflight and artificial gravity on the mouse retina: A progress report." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Gifford G, Boerma M, Sridhara V, Nemec-Bakk AS, Mao XW. "Myocardial changes in a mouse model of prolonged spaceflight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024

Abstracts for Journals and Proceedings	Ilangovan H, Hoadley KA, Eley G, Elgart R, Eslami P, Mao XW. "Multi-omics study of the effect of redox-active metalloporphyrin on murine retina during spaceflight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Seidler RD, Mao XW, Tays GD, Wang T, Zu Eulenburg P. "Effects of spaceflight on the brain." Lancet Neurol. 2024 Aug;23(8):826-35. http://dx.doi.org/10.1016/S1474-4422(24)00224-2 ; PMID: 38945144 , Aug-2024
Articles in Peer-reviewed Journals	Houerbi N, Kim J, Overbey EG, Batra R, Schweickart A, Patras L, Lucotti S, Ryon KA, Najjar D, Meydan C, Damle N, Chin C, Narayanan SA, Guarnieri JW, Widjaja G, Beheshti A, Tobias G, Vatter F, Hirschberg JW, Kleinman A, Afshin EE, MacKay M, Chen Q, Miller D, Gajadhar AS, Williamson L, Tandel P, Yang Q, Chu J, Benz R, Siddiqui A, Hornburg D, Gross S, Shirah B, Krumsiek J, Mateus J, Mao X, Matei I, Mason CE. "Secretome profiling reveals acute changes in oxidative stress, brain homeostasis, and coagulation following short-duration spaceflight." Nat Commun. 2024 Jun 11;15(1):4862. https://doi.org/10.1038/s41467-024-48841-w ; PMID: 38862464; PMCID: PMC11166969 , Jun-2024
Articles in Peer-reviewed Journals	Waisberg E, Ong J, Masalkhi M, Mao XW, Beheshti A, Lee AG. "Mitochondrial dysfunction in Spaceflight Associated Neuro-Ocular Syndrome (SANS): A molecular hypothesis in pathogenesis." Eye (Lond). 2024 Jun;38(8):1409-11. https://doi.org/10.1038/s41433-024-02951-3 ; PMID: 38326485; PMCID: PMC11126720 , Jun-2024
Articles in Peer-reviewed Journals	Alwood JS, Mulavara AP, Iyer J, Mhatre SD, Rosi S, Shelhamer M, Davis C, Jones CW, Mao XW, Desai RI, Whitmire AM. "Circuits and biomarkers of the central nervous system relating to astronaut performance: summary report for a NASA-sponsored technical interchange meeting." Life (Basel). 2023 Aug 31;13(9):1852. https://doi.org/10.3390/life13091852 ; PMID: 37763256; PMCID: PMC10532466 , Aug-2023